Minimum Detectable Effect

2026-04-17

Introduction

When the sample size for a study is constrained by budget, time, or availability of participants, the natural planning question is no longer "what n achieves target power?" but instead "given my fixed n, what is the smallest effect I can reasonably hope to detect?" The minimum detectable effect (MDE) reverses the usual sample-size calculation, solving for the effect size at which power equals the target — typically 80 % — given the fixed n, α, and analysis. Reporting an MDE is increasingly required for grant applications with constrained sample sizes, for pilot and exploratory studies, and for honest power-analysis reporting where a study cannot afford to enrol enough participants for the original target.

Prerequisites

A working understanding of statistical power, the relationship between α, effect size, sample size, and power, and the standard test-specific effect-size measures (Cohen's d, r, f^2, w).

Theory

Power, effect size, significance level, and sample size are linked by the non-central distribution of the test statistic, and solving for any one of these quantities given the other three is the standard inversion in any power-analysis framework. The MDE is defined as the effect size at which the planned test achieves a specified power (usually 0.80) under the planned α and the available sample size. Plotted as a curve of MDE vs. n, this gives a clear picture of the design's sensitivity across resource scenarios.

Assumptions

The same assumptions as the underlying test apply (Normality for t-tests, bivariate Normality for correlation, large expected counts for chi-squared, etc.); the MDE is a property of the planned analysis under the design, not a property of the observed data after the fact.

R Implementation

```r
library(pwr)

pwr.t.test(n = 40, sig.level = 0.05, power = 0.80,
           type = "two.sample", d = NULL)

pwr.r.test(n = 100, sig.level = 0.05, power = 0.80)

n_grid <- seq(10, 200, by = 5)
d_grid <- sapply(n_grid, function(n)
  pwr.t.test(n = n, sig.level = 0.05, power = 0.80, type = "two.sample")$d)

plot(n_grid, d_grid, type = "l", lwd = 2, col = "#2A9D8F",
     xlab = "n per group", ylab = "MDE (Cohen's d)",
     main = "Minimum detectable effect at power = 0.80")
```

Output & Results

For a two-sample t-test with $n = 40$ per group, the MDE at 80 % power is $d = 0.64$ (medium-to-large); doubling to $n = 200$ per group reduces the MDE to $d = 0.28$ (small-to-medium). For a correlation test with $n = 100$, the MDE is $r = 0.28$. Plotting the MDE-vs-n curve shows the diminishing-returns relationship that planners should understand when negotiating sample-size constraints.

Interpretation

A reporting sentence: "With the available sample of 40 participants per arm, the minimum detectable standardised effect (Cohen's d) at 80 % power and two-sided $\alpha = 0.05$ is 0.64 — a medium-to-large effect by Cohen's conventions. Effects smaller than this magnitude would likely remain undetected. The smallest clinically meaningful effect for the primary outcome is $d = 0.40$, so the planned study is underpowered for the clinical target; we report the MDE explicitly to clarify the inferential limits and propose a larger follow-up trial." Always discuss MDE relative to the clinically meaningful effect.

Practical Tips

- Report the minimum detectable effect explicitly when sample size is constrained; this prevents inflated claims of null effects ("no significant difference") when the study was simply too small to detect a clinically meaningful effect, a recurring problem in the literature.
- Couple MDE reporting with explicit discussion of whether the MDE is clinically or scientifically meaningful; an MDE that exceeds the smallest meaningful effect tells readers the study could not have detected the truth even if it were present.
- For exploratory pilot studies, report MDE in place of formal hypothesis testing; pilot studies are not powered for inference and the MDE conveys the design's sensitivity honestly.
- MDE inflates rapidly for sub-group analyses (smaller n per cell), cluster-adjusted analyses (larger effective standard errors), repeated-measures designs with low correlation, and any setting where the effective sample size is reduced from the nominal n. Account for these factors in the MDE calculation, not just the headline figure.
- Simulation-based MDE estimation is recommended for complex designs (mixed-effects models, longitudinal data with informative dropout, generalised linear models with heavy-tailed outcomes) where the standard non-central-distribution formulas do not apply directly.
- Pre-register the MDE in the protocol or analysis plan; reporting MDE only after the fact when results turn out non-significant is methodologically suspect, while a pre-registered MDE protects against that interpretation.

R Packages Used

`pwr` family of functions with the effect-size argument set to `NULL` to solve for MDE; `pwrss` for extended MDE calculations across many test types; `WebPower` for an alternative interface; `simr::powerSim()` for simulation-based MDE in mixed-effects designs; `Superpower` for ANOVA-based MDE simulations.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Sample size, power, and Quarto reporting

- Power: closed-form and simulation